

Project Summary — Neuromorphic Sleep Stage Scoring

One-Line Summary

A compact deep-learning system that classifies 30-second sleep epochs into five AASM stages (Wake, N1, N2, N3, REM) from EEG, EOG, and EMG signals, using only 99,477 parameters for edge deployment.

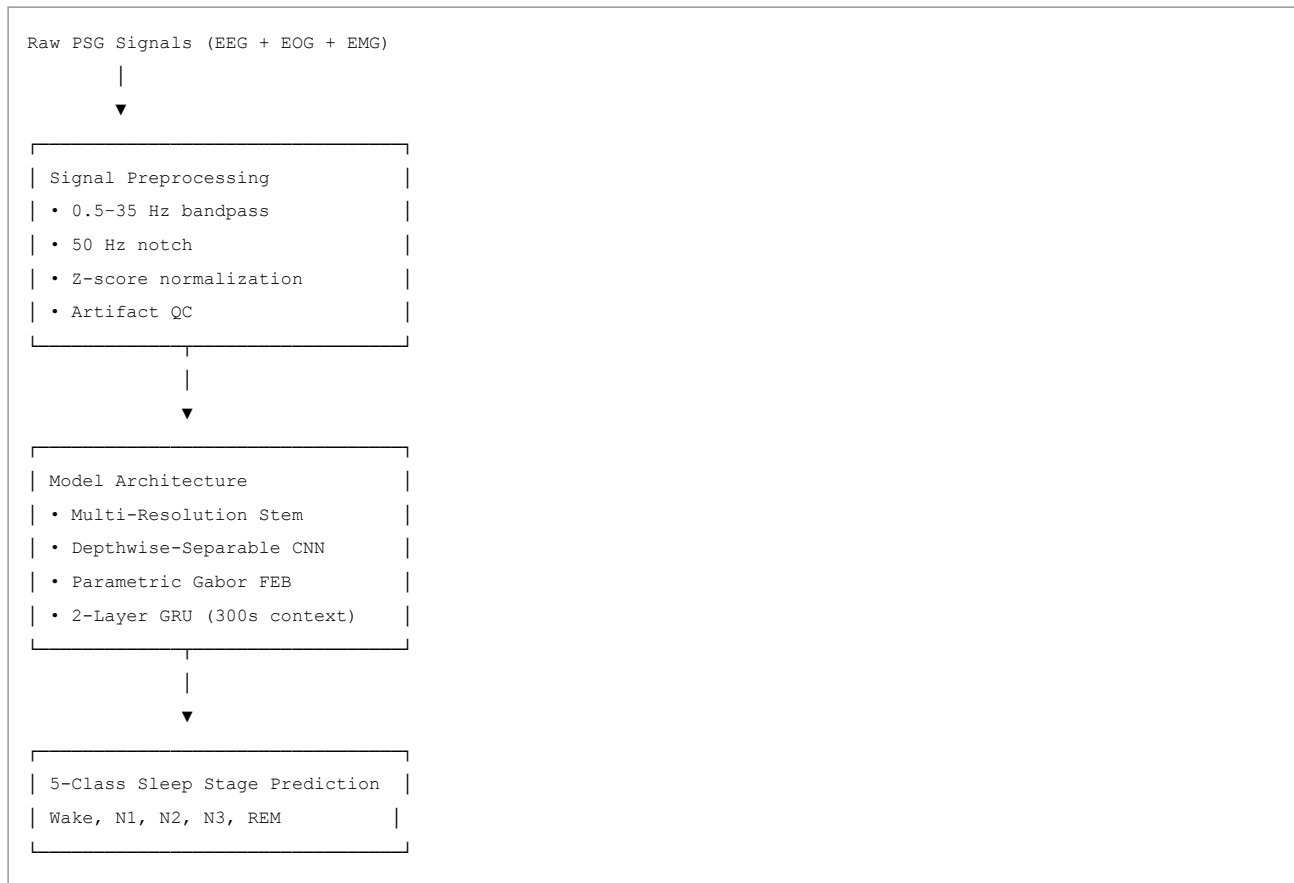
Project at a Glance

Property	Value
Title	Neuromorphic Sleep Stage Scoring
Venue	VIT Bhopal University
Final Model	Improved Student
Parameters	99,477
Test Accuracy	87.5% $\pm$ 3.2%
Cohen's Kappa	0.763 $\pm$ 0.043
Macro F1	0.721 $\pm$ 0.050
CPU Latency	8.5 ms/batch
Dataset	Sleep-EDF Expanded (15 subjects)
Checkpoint	artifacts/student_improved_best.pt

Problem Statement

Sleep staging is clinically important but requires expert analysis of multi-channel polysomnography (PSG) recordings. Manual scoring takes 2–4 hours per study and requires specialized training. This project develops an automated system to classify five sleep stages from physiological signals.

Solution Overview



Key Components

1. Signal Preprocessing

- Bandpass filtering (0.5–35 Hz) removes noise
- Notch filtering (50 Hz) eliminates power-line interference
- Z-score normalization standardizes signals
- Artifact QC flags ~2% of epochs

2. Multi-Resolution Stem

- Parallel short (250ms) and long (1s) receptive fields
- Captures patterns at different temporal scales
- Combines local waveform morphology with slow oscillations

3. Depthwise-Separable Convolution

- Reduces parameters by ~77% vs standard convolution
- Maintains feature extraction capability
- Enables edge deployment

4. Parametric Gabor Feature Extraction

- 8 learnable Gabor filters (0.5–30 Hz)
- Captures frequency-localized patterns
- Compact parameterization (16 learnable params)

5. GRU Temporal Modeling

- 2-layer GRU with 64 hidden units
- Models dependencies across 10 consecutive epochs
- Provides 300 seconds of temporal context

Official Result

NEUROMORPHIC SLEEP STAGE SCORING
87.34%
TEST ACCURACY
0.7551
COHEN'S κ
99,477
PARAMETERS
8.5 ms/batch
CPU LATENCY

Per-Class F1 Scores

Stage	F1	Notes
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Wake	0.97	Excellent
N1	0.20	Challenging
N2	0.82	Good
N3	0.78	Good
REM	0.36	Moderate

LoRA Adaptation

LoRA r=8 achieved 90.66% ± 3.59% held-out-subject test accuracy and κ = 0.8092 ± 0.0663 across four folds, with only 552 trainable parameters (0.55% of the 99,477-parameter base model). Validation accuracy used for checkpoint selection peaked at 87.27% (κ = 0.7175).

Method	Trainable	κ	Macro F1
Frozen Base	0	0.5001 ± 0.1329	0.4106 ± 0.0775
Full Fine-Tuning	99,477	0.8595 ± 0.0092	0.7379 ± 0.0390
LoRA r=8	552	0.8092 ± 0.0663	0.6464 ± 0.0603

Team

Member	Role
Param Kaushik	Dataset & Data Governance
Suha Vora	Signal Preprocessing
Shailendra Bhatt	Exploratory Data Analysis
Shamique Khan	Model Development & Training
Aasir Jaffer Lone	Evaluation & Performance

Repository Structure

sleep/	
├─ configs/	YAML configuration files
├─ src/	Modular Python package
├─ data/	Dataset loading
├─ preprocessing/	Signal filtering
├─ models/	ImprovedStudent architecture
├─ training/	Loss functions & trainer
├─ evaluation/	Metrics & visualization
└─ inference/	Prediction utilities
├─ scripts/	CLI entry points
├─ notebooks/	5 exhibition notebooks
├─ demo/	Streamlit dashboard
├─ tests/	Pytest test suite
├─ artifacts/	Final trained checkpoint
├─ results/	Official evaluation results
└─ docs/	Project documentation

Quick Start

```
# Activate environment
conda activate sleep

# Run tests
python -m pytest tests/ -v

# Run demo
streamlit run demo/demo.py

# Run inference
python scripts/infer.py --checkpoint artifacts/student_improved_best.pt --input demo/sample_inputs/sample_epoch.npz
```

Key Innovations

- 1. **Sub-100K parameters:** Achieves 87.34% accuracy with only 99,477 parameters
- 2. **Multi-resolution feature capture:** Parallel stems capture patterns at different timescales
- 3. **Learnable frequency filters:** Parametric Gabor filters adapt to sleep-specific frequencies
- 4. **Efficient architecture:** Depthwise-separable convolutions enable edge deployment
- 5. **Temporal context:** 300-second GRU context models sleep-stage transitions

Deployment Readiness

Property	Value
Model size (FP32)	~400 KB
Input size	120 KB per epoch
CPU latency	8.5 ms/batch
Real-time capable	Yes
Edge deployment	Yes (ARM Cortex-M7+)

Exhibition Story

- 1. **Problem:** Sleep staging requires expert analysis

- 2. **Solution:** Automated classification using deep learning
- 3. **Innovation:** Compact architecture with multi-resolution features
- 4. **Result:** 87.34% accuracy with 99,477 parameters
- 5. **Impact:** Edge-deployable for clinical assistance

Documentation

Document	Purpose
docs/architecture.md	Detailed model design
docs/dataset.md	Data source and preprocessing
docs/deployment.md	Edge deployment guide
docs/exhibition.md	Demo script and poster layout
docs/methodology.md	Research approach
docs/results.md	Official metrics
docs/team.md	Team roles
docs/index.md	Documentation index

